■ NIntegrate

`NIntegrate[`f`, {`x`, `$xmin$`, `$xmax$`}]` gives a numerical approximation to the integral $\int_{xmin}^{xmax} f\, dx$.

Multidimensional integrals can be specified, as in `Integrate`. ■ `NIntegrate` tests for singularities at the end points of the integration range. ■ `NIntegrate[`f`, {`x`, `x_0`, `x_1`, ..., `x_k`}]` tests for singularities at each of the intermediate points x_i. If there are no singularities, the result is equivalent to an integral from x_0 to x_k. You can use complex numbers x_i to specify an integration contour in the complex plane. ■ The following options can be given in `NIntegrate`:

`AccuracyGoal`	`6`	digits of absolute accuracy sought
`MaxRecursion`	`6`	maximum number of recursive subdivisions
`MinRecursion`	`1`	minimum number of recursive subdivisions
`Points`	`Automatic`	initial number of sample points
`SingularityDepth`	`3`	number of recursive subdivisions before changing variables
`WorkingPrecision`	`Precision[1.]`	the number of digits used in internal computations

■ `NIntegrate` uses an adaptive algorithm, which recursively subdivides the integration region as needed. `Points` specifies the number of initial points to choose in each dimension. `MinRecursion` specifies the minimum number of recursive subdivisions to try. `MaxRecursion` gives the maximum number. ■ You should realize that with sufficiently pathological functions, the algorithms used by `NIntegrate` can give wrong answers. In most cases, you can test the answer by looking at its sensitivity to changes in the setting of options for `NIntegrate`. ■ `N[Integrate[...]]` calls `NIntegrate`. ■ See page 470.

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